Carbon Pricing Relief and Sourcing Choices among Italian Chemical SMEs: Evidence from a Proposed Randomized Controlled Trial of EU ETS Vouchers

Qingyue Chen, Puran Dou, Yi Dou, Gavin Gondalwala, Catherine Morris

Abstract

This Pre-Analysis Plan (PAP) outlines a randomized controlled trial (RCT) designed to investigate the presence of carbon leakage within the Italian chemical manufacturing sector resulting from EU Emissions Trading System (ETS) pricing. As the European Union mandates stricter ETS thresholds and introduces the Carbon Border Adjustment Mechanism (CBAM), Italian chemical firms – particularly small & medium enterprises (SMEs) – face rising compliance costs that may incentivize shifting procurement to jurisdictions with little to no regulation.

The study focuses on firms involved with HS Chapters 28 and 29 (inorganic and organic chemicals) over the period 2026–2030. It includes approximately 1,300 eligible Italian chemical SMEs, which will be randomly assigned to a treatment group receiving a temporary emissions credit or to a control group that does not receive the credit. The credit is worth 4.3% of each treated firm's baseline allotment, effectively offsetting the current Linear Reduction Factor to lower the carbon-cost burden.

Utilizing an OLS regression, this study will measure changes in the share and volume of imports from non-ETS jurisdictions to address the research question: *Does EU ETS carbon pricing cause carbon leakage in Italian chemical manufacturing through upstream supply chain adjustment measured as increased imports of industrial inputs sourced from unregulated suppliers?* By identifying whether targeted financial relief can reverse supply chain-based leakage, this study aims to provide insight into the ongoing tension between decarbonization objectives and industrial competitiveness. Although this may increase domestic emissions, the findings may contribute to future policy efforts aimed at reducing net global emissions as firms substitute away from jurisdictions with little to no regulation.

Table of Contents

1 Introduction

1.1 Background

The EU operates one of the most comprehensive emissions regulatory frameworks in the world, anchored by the Emissions Trading System (ETS). As of 2026, the ETS covers around 40 percent of the EU's greenhouse gas emissions.[1] Since the launch of ETS in 2005, emissions in covered sectors have fallen by 50 percent.[2] While this is a massive win for the climate, some critics argue that it has come at the expense of Europe's economic competitiveness. To investigate this tension, we propose conducting a study of the impact of emissions vouchers on the sourcing choices of Italian chemical firms and carbon leakage. In this context, carbon leakage refers to the relocation of production or sourcing toward jurisdictions with weaker carbon regulations.

1.2 What is the EU ETS?

The ETS operates on a "cap and trade" principle: a cap limits the total amount of greenhouse gases that can be emitted by covered sectors, and that cap declines annually in line with the EU's climate targets.[3] One EU Allowance (EUA) gives the holder the right to emit one tonne of CO_2 equivalent. Put simply, companies receive or buy emission allowances and at the end of each year must surrender enough allowances to cover their actual emissions or face penalties.[4] Enforcement mechanisms include €100 penalties per excess tonne of CO_2.[5] If they emit less, they can sell spare allowances; if they emit more, they must buy extra. This creates a financial incentive to reduce emissions. With EUA prices hovering around €75 per tonne as of April 2026,[6] and chemical production being energy-intensive, these costs are non-trivial, particularly in an era of fluctuating energy prices.

[1] European Commission, “About the EU ETS,” Retrieved March 25, 2026.
[2] European Commission. “EU Emissions Trading System Has Reduced Emissions in the Sectors Covered by 50% Since 2005.” April 4, 2025.
[3] European Commission, “About the EU ETS,” Retrieved March 25, 2026.
[4] Senken, “EU ETS,” 2026.
[5] Offset8 Capital, “EU Carbon Market 2025,” 2025.
[6] Trading Economics, “EU Carbon Permits,” 2026.

Allowances reach firms through two channels. They are primarily sold via auctions, though companies also receive some allowances for free. If an installation reduces its emissions, it can sell spare allowances or bank them for future use.[7] Free allocation has historically served as a carbon leakage protection measure — sectors deemed at risk of relocating production outside the EU receive a higher share of allowances at no cost, dampening the compliance cost burden. However, this share is being phased out: from 2026, free allocation for installations in sectors covered by the Carbon Border Adjustment Mechanism (CBAM) will begin to phase out, shifting the cost burden onto firms more fully.[8]

The system is currently in its fourth trading phase (2021–30), which is the most ambitious to date. The linear reduction factor — the annual rate at which the cap shrinks — increased from 2.2% per year (2021–2023) to 4.3%, and is projected to continue increasing, targeting a 62% reduction in ETS-sector emissions by 2030 versus 2005 levels.[9]

1.3 The European Chemical Industry is Being Hit by High Regulatory and Energy Costs

Europe is a major chemical producer—second behind China.[10] For years, Europe's share of the market has fallen (Figure 1). Observers attribute the current weakness of the European market to high energy prices, worsened lately by the conflict in Iran; stringent environmental regulation, including ETS; and growing competition from Asian producers. One industry survey, published in late 2025, found that investments in the industry fell by a precipitous 86 percent from 2024 to 2025, while plant closures increased sixfold, representing approximately 9 percent of the European chemical production capacity.[11] Nearly half of the companies surveyed cited energy cost competitiveness as the primary reason for plant closures, while 8 percent attributed plant closures to environmental regulations.[12]

[7] European Commission, "About the EU ETS," Retrieved March 25, 2026.
[8] International Carbon Action Partnership, "EU Emissions Trading System ," 2026.
[9] Senken, "EU ETS," 2026.
[10] China surpassed the European Union in market share in 2009. Echemi, "How did the European chemical industry give birth to giants such as BASF/Evonik?," 2020.
[11] Roland Berger, *European Chemical Closures and Investments Radar 2022–2025*, 2025.
[12] Ibid.

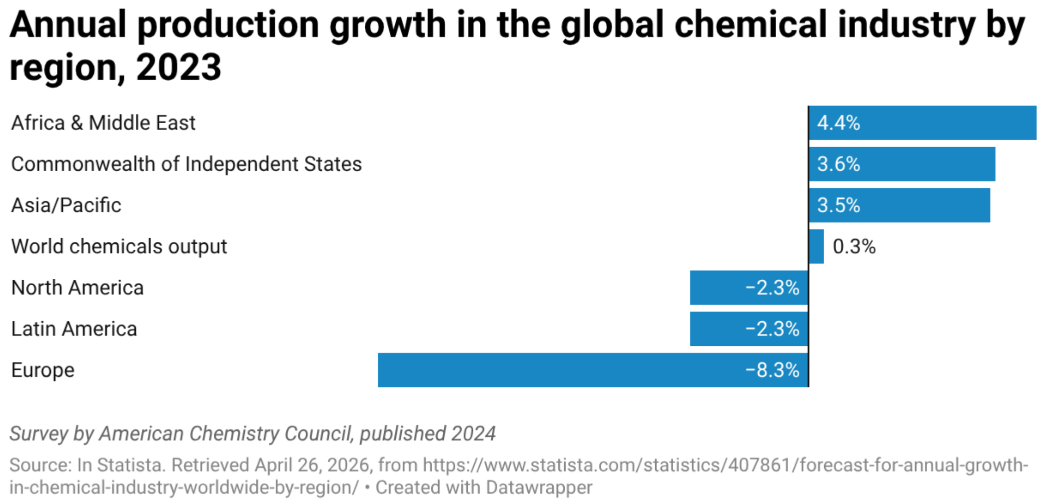

Figure 1. Recent data show slowing production in the European chemical industry.

Trouble in the chemical sector is not just an economic concern, it is also an environmental issue. Chemicals are the “first link” in many production supply chains. When EU-based producers face higher carbon costs than foreign competitors, they may respond by relocating production to jurisdictions without equivalent carbon pricing, or by sourcing intermediate goods from abroad—or simply be undercut by firms in countries with less stringent regulation. This makes rising imports from countries like China particularly concerning: While China’s CO2 emissions were either flat or falling in 2025 across most sectors, the one major exception was in China’s growing chemical industry—emissions in this industry grew by 12 percent in 2025.[13] This “carbon leakage” exports emissions rather than reducing them globally, undermining the environmental effectiveness of the ETS while harming EU industrial competitiveness. As cheaper, carbon-intensive imports from countries without equivalent carbon pricing flood the EU market,[14] European industrial output and employment is being eroded while hindering progress on reducing global emissions.

[13] Myllyvirta, “Analysis,” 2026.
[14] ICN Bureau, “Europe Faces Chemical Industry Collapse As Chinese Imports Flood Market,” 2025.

1.4 The Italian Chemical Industry Offers a Compelling Case Study

Italian chemical firms present a compelling case study for evaluating policy interventions. While Italy is a key producer—the third largest in Europe—it has not been quite as hard hit as countries like Germany or the Netherlands. Plant closures in Italy account for just 7 percent of the overall closures.[15] Thus it offers comparative stability for research purposes. Additionally, there are approximately 3,900 chemical manufacturing firms in the country,[16] providing a sufficient sample for research purposes. Much of chemical production is centered in the industrial north, particularly in the region of Lombardy.[17] However, the sector is currently being squeezed both by high carbon costs and high energy costs. Since a large share of Italian firms are small and medium-sized enterprises (SMEs), they have less capacity to absorb price volatility.[18]

Italy perceives rising costs as an urgent issue, and, leading an eleven-member coalition of EU member states known as the "Friends of Industry," has called for a temporary suspension of the ETS pending comprehensive reform, arguing that the system functions as a *de facto* tax on European industry and undermines international competitiveness.

The chemical industry is a key concern. Italian Industry minister Adolfo Urso has said that the ETS has a "perverse effect" on European competitiveness, adding, "If we are in the face of the collapse of the European chemical industry and the crisis of European ideology, we cannot wait for the time of negotiations within the European Union to find a solution."[19] Italy's position is contested within the EU: eight member states, including Denmark, Sweden, and the Netherlands, have formally opposed suspension, describing the ETS as the cornerstone of EU climate policy.

Proponents of ETS argue that the true culprit behind prohibitive prices is the EU's continued dependence on fossil fuels[20]—an issue currently exacerbated by the war in Iran. While recent events have heightened this debate, at a fundamental level, it reflects a genuine tension

---

[15] Roland Berger, *European Chemical Closures and Investments Radar 2022–2025*, 2025.
[16] Eurostat. (May 13, 2025). Number of chemicals and chemical products manufacturing firms in Italy from 2021 to 2024 (in 1,000s). In *Statista*. 2026.
[17] Cefic, "Landscape of the European Chemical Industry," 2026.
[18] The average firm has 55 employees, according to industry sources. Federchimica Confindustria, *The Chemical Industry in Italy,* 2025; Federchimica Confindustria, *The Role of Foreign-Owned Chemical Companies in Italy*, 2019.
[19] Pacheco, "Italy calls for suspension of EU carbon market," 2026.
[20] Elderson, "Europe's fossil fuel dependence poses risks to price stability," 2026.

between decarbonization objectives and industrial competitiveness. Even the European Commission acknowledges that the expansion of environmental and chemical regulations may outpace firms' ability to adapt—which could weaken their economic competitiveness, and force them to cede market share to firms in countries with less stringent regulations[21]—as is currently occurring in the chemical sector.

1.5 Regulatory Hope on the Horizon

ETS creates a cost wedge that disadvantages domestic products relative to imports that is currently not being addressed by an equivalent tax on imports. This is key because ETS compliance costs apply to *domestic production only*. Italian firms that emit CO_2 during manufacturing must either hold sufficient allowances or purchase them on the market. Thus, when a firm imports an intermediate chemical good from a supplier in China or India, the emissions embedded in that good's production are not captured by the EU ETS. This asymmetry is at the heart of the carbon leakage concern—and of this study's research design.

However, there is a regulatory solution to this asymmetry on the horizon. The Carbon Border Adjustment Mechanism (CBAM), phasing in from 2026, is designed to address this very issue by imposing a carbon price on imports of certain goods, while also protecting European business interests. However, CBAM currently covers only a narrow set of sectors (steel, cement, aluminum, fertilizers, electricity, and hydrogen) and does not extend to the chemical intermediates under HS Chapters 28 and 29 that are central to this study.[22] To date, imported chemical inputs from third countries remain effectively carbon-cost-free. However, the EU is considering extending the CBAM to chemicals in 2030,[23] and given the challenges that the industry faces, may introduce them even sooner.[24]

[21] European Committee of the Regions. A Place-Based Approach to Strengthening the EU Chemicals Industry in Regions and Cities. Commission for Economic Policy, 2026.
[22] European Commission, "CBAM Sectors," 2026.
[23] Minten et al., "Embodied Emissions of Chemicals within the EU Carbon Border Adjustment Mechanism," 2025; and Thompson, "The Carbon Border Adjustment Mechanism (CBAM): Overview and Timeline," 2025.
[24] Kuppel, "CBAM Timeline, Deadlines, and Phases," 2026.

2 Literature Review

The empirical literature on the EU ETS and firm behavior can be organized into three overlapping strands: (1) the system's effectiveness in reducing emissions and effects on firm productivity and competitiveness; (2) whether and to what extent it has induced innovation and clean technology investment; and (3) whether it has caused carbon leakage through production relocation or supply chain adaptation. These debates are phase-dependent and method-dependent, and the early "null result" consensus has been substantially revised as the policy has become more stringent. As noted in the introduction, the ETS program is currently in its fourth and most stringent phase.

2.1 Emissions Abatement and Firm Performance

The dominant finding of the early empirical literature is that the EU ETS reduced emissions without imposing measurable costs on firm competitiveness. Verde (2020), in a comprehensive survey of evidence through Phases I and II (2005–2012), concludes there is no evidence of widespread negative effects on competitiveness or significant carbon leakage.[25] This null result reflected over-allocation of free allowances and extensive cost pass-through to consumers rather than genuine insulation from compliance pressure. Naegele and Zaklan (2019) study EU manufacturing trade flows from 2004 to 2011 and find no evidence of leakage during this period,[26] consistent with the Verde (2020) review. The standard explanation was free allocation — sectors at leakage risk received allowances at no cost, dampening the competitive asymmetry.

More recent work drawing on Phase III data has begun to revise this picture. Colmer, Martin, Muûls, and Wagner (2024), in the *Review of Economic Studies*, find that the EU ETS induced regulated French manufacturing firms to reduce CO_2 emissions by 14–16 percent with no detectable contractions in economic activity — firms made targeted investments in energy-saving capital.[27] Importantly, however, this result was established for regulated firms making capital investments, not for downstream buyers of regulated inputs.

[25] Verde, "The Impact of the EU Emissions Trading System on Competitiveness and Carbon Leakage," 2020.
[26] Naegele and Zaklan, "Does the EU ETS Cause Carbon Leakage in European Manufacturing," 2019.
[27] Colmer et al., "Does Pricing Carbon Mitigate Climate Change?," 2024.

2.2 Supply Chain Evidence

More recent work has produced substantially stronger evidence of carbon leakage, particularly through a channel earlier studies missed: supply chain adaptation. Coster, di Giovanni, and Mejean, (Federal Reserve Bank of New York, 2024) study French firm-level import data and find that French firms increasingly shifted imports of ETS-regulated inputs to non-EU countries as the policy became more stringent. The share of ETS-regulated products sourced from outside the EU rose by 4.3 percentage points after ETS implementation.[28]

Furthemore, Böning, Di Nino, and Folger (ECB Working Paper, 2023) provide firm-level evidence showing that the relative share of "dirty" inputs from non-ETS countries began increasing in Phase I (2005–2007) and became significantly positive by the end of Phase II. The pattern is consistent with firms adapting sourcing strategies to minimize carbon policy costs — carbon leakage through the value chain rather than production relocation.[29]

Finally, Minten et al. (2025) argue that the current CBAM framework would only capture 50–60% of the production emissions embedded in those chemicals. CBAM is designed to price emissions at the point of manufacture, but chemical production relies heavily on fossil feedstocks and refinery products that generate emissions upstream of the manufacturing step itself. Those upstream emissions fall outside CBAM's scope.[30]

2.3 Gaps This Study Addresses

The existing literature leaves three gaps this study is positioned to fill. First, most evidence covers Phases I and II, when prices were low and free allocation was generous. Phase IV— with prices hovering between €60-90, a tightening Linear Reduction Factor, and free allocation phase-out for CBAM-covered sectors — is structurally different. Second, the literature has largely neglected SMEs; as Verde (2020) notes, firm size effects need further investigation, and Italian chemical SMEs are precisely the context where size-differentiated effects are most likely.

[28] Coster, Pierre, Julian di Giovanni, and Isabelle Mejean. "Firms' Supply Chain Adaptation to Carbon Taxes," 2025.
[29] Böning et al., "Benefits and Costs of the ETS in the EU, a Lesson Learned for the CBAM Design," 2023.
[30] Minten et al., "Embodied Emissions of Chemicals within the EU Carbon Border Adjustment Mechanism," 2025.

3 Motivation

China's threat to European chemical firms operates through two distinct channels: (1) as a competitor undercutting European producers of final/commodity chemicals on price, and (2) as an emerging supplier of intermediates in specific sub-segments. We propose to evaluate the second channel. CEFIC data confirm that Europe's upstream chemical segments—basic chemicals and feedstocks—have been in trade deficit since 2011, reflecting dependence on imported raw materials.[31] And while China initially built its chemical industry around basic chemicals, it is now moving up the value chain into intermediate and specialty chemicals.[32]

The Italian chemical sector provides an unusually clean setting in which to study the relationship between ETS compliance costs and intermediate goods sourcing. Production in the broader chemicals sector—classified under HS Chapters 28 and 29—is subject to ETS obligations for covered installations, embedding carbon costs into the price of domestically produced intermediates.

This configuration creates a testable mechanism. ETS compliance costs raise the effective cost of domestic production, potentially making imported intermediates from lower-cost, higher-emissions suppliers relatively more attractive. Indeed, the existing literature shows that carbon pricing meaningfully shapes global supply chain composition.

If firms are responding to ETS cost pressure by shifting sourcing toward less regulated suppliers, the environmental benefit of the regulation is partially or fully offset by upstream emissions abroad. Conversely, if ETS cost relief—modeled here as an emissions credit—induces firms to shift sourcing back toward EU-based suppliers, it opens up the possibility of investigating the following questions:

- Does carbon leakage exist in the current market structure and is it being exacerbated by market pressures?
- Can a targeted, temporary ETS subsidy reverse leakage at the firm level?
- Additionally, how significant does ETS cost relief need to be in order to meaningfully shift sourcing? This question has important implications for a sector currently under pressure.

[31] David, Sébastien, Christian Popa, Gunter Lipowsky, Paul Petrescu et al. (Advancy). The Competitiveness of the European Chemical Industry. CEFIC, January 2025.
[32] Grimes, "China's Evolving Role in the Chemical Global Value Chain," 2023

In the course of studying these questions, the study will also provide valuable insight into the Italian chemical sector. Because this study focuses on the Italian chemical sector's sourcing of intermediate goods during the period 2020–29, in the leadup to 2030 (when chemicals are expected to be covered under CBAM) it will also provide insight into whether firms change their behavior in anticipation of the next regulatory phase.

4 Theory of Change

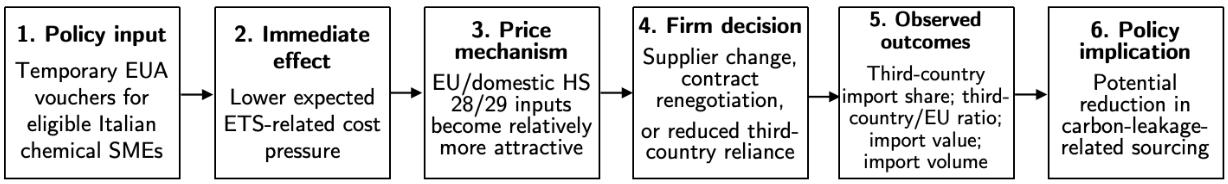

Figure 2. Theory of Change

4.1 EU ETS Emissions Credit and Sourcing Behavior of Italian Chemical Firms

The Italian government will provide approximately 650 Italian chemical firms with a temporary emissions credit worth 4.3% of their baseline EUA allotment; this value offsets the current ETS Linear Reduction Factor of 4.3%, with each voucher valued at approximately EUR 73.43.[33] With average emissions at 625 tonnes CO2 eq., firms will receive an average of 26.9 credits, for an annual cost of approximately EUR 1.3 million.[34] By providing vouchers, firms should be incentivized to alter sourcing away from emissions intensive third countries, which they could not have done before without additional allowances. Without cost relief, firms are constrained to “clean” EU-based and domestically produced inputs which may be relatively more expensive because ETS compliance costs are embedded in production costs. Firms may therefore rely more heavily on lower-cost third-country suppliers. The EUA voucher is expected to reduce this cost pressure and make EU-based or domestic sourcing relatively more attractive. The intervention assumes that ETS compliance costs are partially embedded in the relative price of EU-produced intermediates, such that reducing compliance costs changes sourcing incentives at the margin.

Firms in the treatment group, facing lower carbon cost burdens, find that EU-sourced intermediates become relatively more attractive compared to cheaper, carbon-intensive imports from China and other third countries. As a result, treated firms shift their procurement of HS28/29 chemical intermediates toward EU-based suppliers and away from unregulated ones. At

[33] International Carbon Action Partnership, “EU Emissions Trading System ,” 2026.
[34] EU Emissions Tracker. (n.d.). *Chemical industry*. Retrieved May 1, 2026, from https://emission-tracker.eu/italy/industry/chemical-industry

endline, the study compares sourcing composition between treated and control firms, identifying the causal effect of ETS cost relief on supply chain decisions.

As such, this study aims to address the research question:

Does EU ETS carbon pricing cause carbon leakage in Italian chemical manufacturing through upstream supply chain adjustment measured as increased imports of industrial inputs sourced from unregulated suppliers?

4.2 Timeline

The study period runs from 2026 to 2030, with four years of treatment from 2026 through 2029. Because supplier relationships and procurement contracts may take multiple years to adjust, a four-year treatment window allows the study to observe sourcing changes that may not occur immediately after the first voucher distribution.

- 2026 Q1–Q2: Screen eligible firms, recruit participants, and collect baseline data.
- 2026 Q3: Conduct random assignment and distribute the first EUA vouchers.
- 2026 Q4–2029 Q4: Issue annual EUA vouchers and conduct quarterly procurement surveys.
- 2030 Q1–Q2: Conduct endline surveys, verify EUA serial numbers, and match baseline, endline, and administrative data where available.
- 2030 Q3–Q4: Analyze data and report results.

4.3 Cost

The direct voucher cost is calculated as follows:

Assumptions:

- Target sample size = 1,300 Italian chemical SMEs
- Treatment-control assignment = 50/50
- Number of treated firms = 650 Italian chemical SMEs
- Average baseline emissions per firm = 625 tonnes CO_2e per year
- Voucher size = 4.3% of baseline EUA needs
- EUA price = EUR 73.43 per EUA
- Study period = 2026–2030

- Treatment period = 4 years, from 2026 through 2029

Step 1. Calculate EUAs per firm per year

EUAs per firm per year = average baseline emissions per firm × voucher size

= 625 × 0.043 = 26.875 EUAs per firm per year ≈ 26.9 EUAs per firm per year

Step 2. Calculate the annual voucher value per firm

Annual voucher value per firm = EUAs per firm per year × EUA price

= 26.875 × 73.43 = EUR 1,973.43 ≈ EUR 1,973 per firm per year

Step 3. Calculate the annual direct voucher cost for treated firms

Annual direct voucher cost = annual voucher value per firm × number of firms

= EUR 1,973.43 × 650 = EUR 1,282,729.50 ≈ EUR 1.3 million per year

Step 4. Calculate the total direct voucher cost over four treatment years

Total direct voucher cost = annual direct voucher cost × number of treatment years

= EUR 1,282,730 × 4 = EUR 5,130,921 ≈ EUR 5.1 million

Therefore, under the SME-only design with approximately 1,300 total participating firms and balanced treatment-control assignment, the direct voucher cost would be approximately EUR 1.3 million per year, or approximately EUR 5.1 million over four years of treatment. This estimate excludes survey administration, compliance verification, data matching, and research implementation costs.

5 Experimental Design

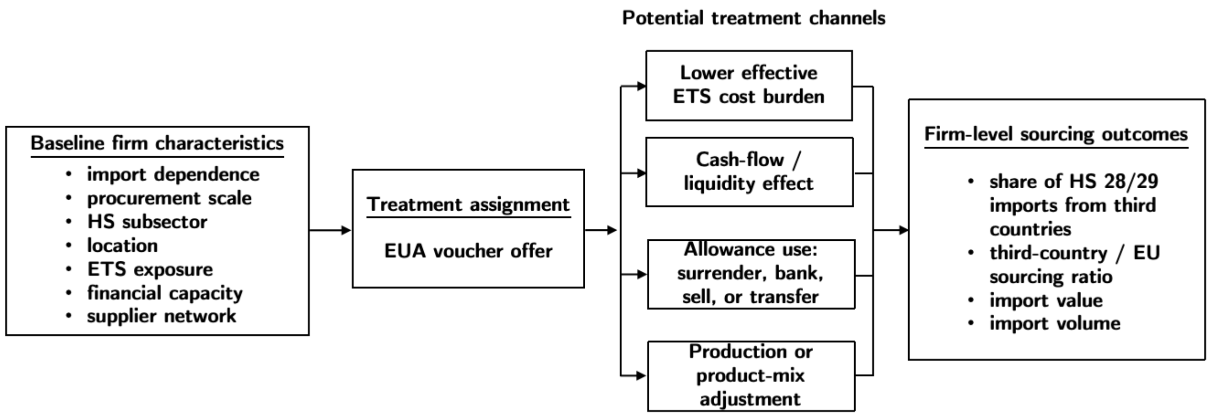

Figure 3. Causal DAG

5.1 Intervention/Manipulation

The intervention is a temporary emissions credit provided to eligible Italian chemical firms. Each treated firm receives a voucher equivalent to 4.3% of its baseline EUA allotment, a value calibrated to offset the current EU ETS Linear Reduction Factor, with each credit valued at approximately €73.43. Based on average installation-level emissions of 625 tonnes CO2 equivalent, treated firms will receive an average of 26.9 credits per year. Participation in the program is voluntary. The credit is designed to reduce the effective compliance costs associated with the EU ETS, thereby lowering the carbon-cost burden embedded in firms' production and sourcing decisions. Without cost relief, firms face pressure to rely on lower-cost third-country suppliers because ETS compliance costs are embedded in the price of EU-based and domestically produced inputs. The voucher is expected to reduce this cost pressure and make EU-based sourcing relatively more attractive. The purpose of the intervention is therefore to test whether targeted ETS cost relief induces firms to shift procurement of HS 28/29 chemical intermediates toward EU-based suppliers and away from unregulated third-country ones.

5.2 Enrollment Criteria/Sampling Strategy

The study population is drawn from Italian chemical firms producing or importing products classified under HS Chapters 28 and 29. Federchimica Confindustria categorizes firms in this sector into three strata: foreign-owned companies, which account for 37% of total production value; medium- and large-sized Italian groups, defined as those with worldwide sales exceeding €100 million, which account for 23%; and Italian SMEs, which account for the remaining 40%. To be included in the sample, firms must have observable purchasing or import activity in intermediate chemical goods during the baseline period, must remain active throughout the study period, and must have a sufficient baseline scale of procurement to allow detection of changes in sourcing patterns.

This study focuses exclusively on the Italian SME stratum, targeting approximately 1,300 firms. This choice is motivated by two considerations. First, from a policy perspective, SMEs are the segment most likely to face binding cost constraints: with an average of 55 employees and limited financial buffers, they have less capacity than larger firms to absorb ETS-related price volatility, making them the most policy-relevant group for evaluating targeted cost relief. Second, from an operational perspective, the average voucher cost per SME is substantially lower than for the other two strata, since SMEs have lower baseline import volumes and therefore lower ETS-attributable compliance costs, making the intervention fiscally feasible at the required sample size. Within the SME stratum, firms will be randomly selected from the eligible population prior to assignment to treatment and control conditions.

5.3 Assignment Protocol/Identification Strategy

After baseline data collection, eligible firms will be randomly assigned to either a treatment group that receives the emissions credit or a control group that does not. Random assignment ensures that, on average, treated and control firms are comparable in both observed and unobserved characteristics prior to the intervention. The causal effect of the emissions credit will be identified by comparing post-intervention sourcing outcomes between the two groups. The main outcomes include the share of intermediate imports sourced from unregulated countries, the ratio of imports from third countries to EU-ETS countries, and the total volume of imported intermediate chemical goods. If the experiment is properly randomized, the post-treatment difference between treated and control firms can be interpreted as the causal effect of reducing effective ETS compliance costs.

Since firms are operating on a unified market, however, there is a risk of contamination of control firms purchasing credits from treatment firms. This could create a downward bias in our effects due to control firms also shifting their sourcing behavior. While this does reflect the reality of the market, the study will utilize EUA serial numbers tracked on the Union Registry in the endline survey for a second regression. If any control firms purchased EUAs released as part of the experiment, they will be excluded from the second sample, and both sets of results will be reported.

6 Data

6.1 Inputs, Outputs and Outcomes of Interest

In this study, we focus on the Italian chemical sector, defined as products classified under HS Chapters 28 and 29 (inorganic and organic chemicals). We conceptualize firm behavior within a production and sourcing framework, where firms combine intermediate inputs to produce output under changing carbon cost conditions.

Inputs are proxied by imported intermediate chemical goods, measured using trade data at the HS6 level for HS28–29. We restrict the sample to Italian imports and interpret these flows as intermediate inputs into domestic chemical production. Input measures include total import value as well as the distribution of imports across these partner groups. To capture sourcing patterns, we classify partner countries into two groups: EU-ETS-Covered Suppliers and Third countries:

EU-ETS-Covered Suppliers	Third countries
All jurisdictions are covered by EU ETS for the chemical sector	*Jurisdictions where there is no national ETS, or where the chemical sector is excluded*
EU Countries	Such As: China India United States

Outputs refer to production-related outcomes in the Italian chemical sector. Given the absence of firm-level production data, we proxy output using export values of HS28–29 chemical products from Italy, which reflect changes in production scale and international competitiveness. This allows us to assess whether changes in input costs and sourcing patterns translate into observable shifts in sectoral output. Our endline survey targets these outputs by assessing if firms used vouchers for emissions-intensive inputs as intended, or sold vouchers by tracking EUA serial numbers.

Outcomes of interest capture changes in sourcing behavior consistent with carbon leakage. The main outcome variables include:

(1) the share of imports sourced from third countries,

(2) the ratio of imports from third countries to EU-ETS countries, and

(3) the total volume of imported chemical inputs.

6.2 Data Collection Strategy & Data Sources

There will be a baseline survey issued to firms at the beginning of the study which collects information on their current sourcing partners, with an endline survey at the end to assess behavior changes. The survey is administered to the firm's Procurement or Supply Chain Manager. A total of 8 questions are collected across the two waves, organized as follows.

<u>Baseline Only</u>

Firm Profile

- Q1: How would you classify your firm? *(Foreign-owned / medium- or large-Italian group / Italian SME)*
- Q2: What is your annual procurement spend on HS 28/29 intermediates? *(<€1M / €1–10M / €10–50M / >€50M)*

ETS Cost Exposure

- Q5: What was your facility's total EUA surrender for domestic manufacturing operations during the last fiscal year? *(Numeric, tonnes CO_2e)*
- Q6: What were the estimated EU CBAM certificate costs associated with your imports last fiscal year? *(None / <€50K / €50–500K / >€500K)*

<u>Baseline + Endline</u>

Sourcing Behavior

- Q3: What share of your HS 28/29 inputs were sourced from each region? *(Must sum to 100%: Italy / Other EU / China / India / Other non-ETS)*
- Q4: What are the primary reasons for choosing your current sourcing partners? *(Select up to 3: Price / Quality / Long-term contract / Proximity and lead time / Compliance / Limited EU supply)*

Endline Only

Behavioral Change

- Q7: Compared to 12 months ago, how has sourcing from EU-based suppliers changed? *(Increased significantly / Increased somewhat / No change / Decreased)*
- Q8: If ETS costs were reduced, how likely would you be to shift procurement toward EU suppliers? *(Likert scale 1–5: Unlikely to Likely)*

Endline Addition (Assignment Protocol)

At the endline, firms will also be asked to report the serial number of any EUAs purchased or used during the study period. This allows the study to identify non-compliance through the misuse or sale of vouchers. If any control firms are found to have purchased EUAs released as part of the experiment via the Union Registry, they will be excluded from the second regression sample, and both sets of results will be reported.

6.3 Hypotheses and Outcomes

The study's working hypothesis is that by working with Italian regional governments to secure EUA vouchers and provisioning them to firms in the treatment group, the effective marginal cost of domestic and EU-based production will be reduced–ultimately incentivizing firms to substitute away from carbon-intensive, non-ETS sourcing. To evaluate this causal mechanism, the study will test the following hypotheses:

Outcome 1: Sourcing Composition/Leakage Reversal

H0: The provision of an emissions credit has no effect on the share of intermediate chemical goods sourced from unregulated suppliers.

H1: Firms receiving the credit will alter the share of their total intermediate imports sourced from unregulated suppliers compared to the control group.

Data construction: Calculated as total import value from third countries divided by total import value from all partner countries.

Outcome 2: Emissions Volume

H0: The provision of an emissions credit has no effect on the total environmental emissions.

H1: Firms receiving the credit will decrease their total volume of imported intermediate goods, increasing domestic emissions but reducing carbon leakage and net global emissions.

Data construction: Calculated as the aggregate volume of all Schedule 28 and 29 substances imported by the firm over the study period, multiplied using the US EPA's Greenhouse Gas Equivalencies Calculator and Climatiq's guidelines for CBAM emissions calculations.[35]

While aggregate macroeconomic shocks — such as rising global energy costs — may impact the sector's overall sourcing behavior, the randomized control design ensures these shocks are distributed symmetrically across the treatment and control groups, protecting the internal validity of the Average Treatment Effect (ATE).

Assessment

The outcomes will be assessed by estimating the following OLS regression equation:

$$Y_i = \beta_0 + \beta_1 Treatment_i + \gamma \mathbf{X}_i + \epsilon_i$$

Where terms are defined as:

$Treatment_i$: A binary indicator variable that equals *1* if firm *i* was randomly assigned to receive the EUA voucher, and *0* if they remained in the control group.

[35] Climatiq, "How to Calculate Emissions for CBAM Reporting," Climatiq (blog), December 29, 2025, https://www.climatiq.io/blog/how-to-calculate-emissions-for-cbam-reporting.

β_1: Coefficient of interest, the causal effect of the emissions credit.

X_i: Vector of strictly pre-treatment covariates (baseline revenue, historical pre-treatment import volumes, or sub-sector indicators), used to increase the precision of estimates

ε_i: Error term

7 Power Calculations

7.1 Parameters

The power analysis is calibrated using the following parameters, with sources noted where applicable.

Stratum-level mean import values (EUR per firm per year), derived from *The Chemical Industry in Italy*, which reports total chemical-sector imports of €53.1 billion across 2,849 firms in 2024[36]:

- Foreign-owned multinationals: €86.1 million
- Medium- and large-sized Italian Groups: €57.7 million
- Italian SMEs: €19.9 million

These per-firm means are obtained by allocating the industry-wide €53.1 billion import total across the three Federchimica strata using import-share weights of 60% / 25% / 15% (Foreign-owned / Med-Large Italian / SME). These weights extend Federchimica's reported finding that foreign-owned firms in Italy export approximately 70% of their production value[37] — a pattern that, in the trade-economics literature, typically co-occurs with proportionally higher import intensity, since export-oriented production relies on internationally diversified supply chains.

Within-stratum dispersion (σ on the log scale), capturing heterogeneity in firm size and procurement scale:

- Foreign-owned: $\sigma_{log} = 0.8$
- Medium- and large-sized Italian: $\sigma_{log} = 1.0$
- Italian SME: $\sigma_{log} = 1.4$

These values are project-specific calibrations rather than estimates from external data. They reflect the stylized fact that foreign-owned multinationals in Italy operate under centralized parent-level procurement systems and are therefore relatively homogeneous in scale, whereas the SME stratum spans a much wider range — from micro-firms with a handful of employees to

[36] Federchimica, "The Chemical Industry in Italy", 2025.
[37] Federchimica, "The Chemical Industry in Italy", 2025.

mid-size producers approaching the €100 million threshold that defines the medium- and large-sized category.

Treatment effect: a 20% reduction in firm-level annual import value, modeled multiplicatively rather than as a fixed absolute amount.

Statistical thresholds: significance level $\alpha = 0.05$ (two-sided) and target power $1 - \beta = 0.80$, following the conventions adopted in our initial proposal.

Sample-size grid and simulation depth: power is evaluated at total sample sizes of 100, 300, 500, …, 1,900 firms per stratum (balanced 50/50 between treatment and control), with 1,000 Monte Carlo iterations at each grid point.

7.2 Assumptions

The simulation rests on four assumptions, each made explicit so that readers can assess the robustness of our sample-size estimates:

- Lognormal import distribution within each stratum: firm-level trade values are right-skewed in essentially all empirical settings, and the lognormal is the standard parametric choice for this distributional shape. Combined with the per-stratum mean and σ_log specified above, this fully determines the distribution of the outcome variable.
- Multiplicative treatment effect: the EUA voucher is assumed to reduce each firm's import value by a constant proportion (20%) rather than by a constant absolute amount. This is consistent with the economic logic of the intervention — a firm facing lower effective ETS compliance costs should adjust its sourcing decisions in proportion to its baseline scale, not by a uniform euro amount.
- Independence across firms: treatment and control firms do not affect each other's outcomes. We acknowledge in the limitations section that this assumption could be violated through supplier-level spillovers (e.g., a treated firm's reduced demand for imports affecting the prices its suppliers charge to control firms), but absent firm-level network data we proceed under the standard SUTVA framework.
- Heterogeneity ranking across strata (Foreign-owned < medium- and large-sized Italian < Italian SME): assumed to reflect greater within-group size dispersion in the SME segment, as discussed above. While this study will solely focus on Italian SMEs, this assumption remains central to future analysis.

7.3 Procedure and Results

For each (stratum, sample size) combination on the grid, we simulate 1,000 RCTs as follows. We draw N/2 control-arm and N/2 treatment-arm firms from the lognormal distribution implied by the stratum's parameters, with the treatment-arm geometric mean reduced by a factor of 0.80 to embed the 20% effect. We then perform a two-sample t-test on the log-transformed import values and record whether the null is rejected at $\alpha = 0.05$. Power is the rejection rate across the 1,000 iterations. Working on the log scale is appropriate here because a multiplicative effect on a lognormal outcome translates into an additive shift on the log scale, where the t-test is well-specified and most powerful.

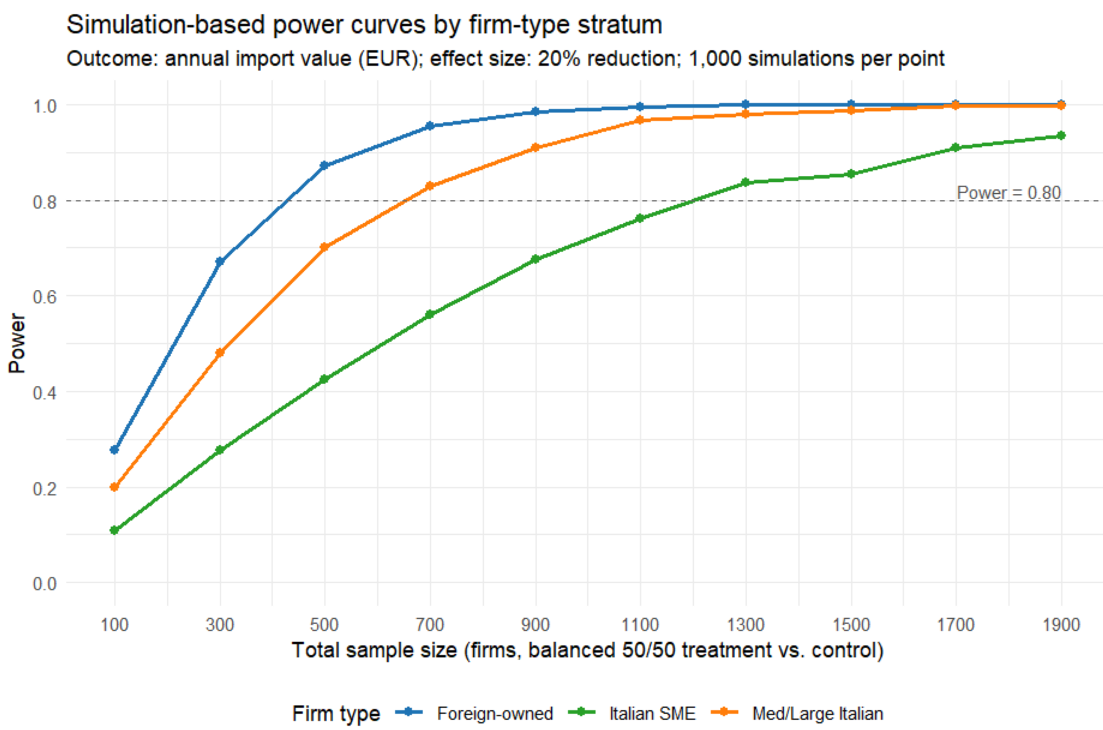

Figure 4. Italian SMEs require the largest sample because of the highest variance

Figure 4 presents the resulting power curves. As expected, power rises monotonically with sample size in all three strata, but the rate of increase differs systematically: Foreign-owned firms reach the conventional 0.80 threshold at the smallest sample size, followed by medium- and large-sized Italian groups, with Italian SMEs requiring the largest sample. This ordering follows directly from the within-stratum dispersion assumption — the higher the variance in baseline imports, the larger the sample required to detect a fixed proportional effect.

Reading the smallest grid point at which simulated power meets or exceeds 0.80, the minimum required sample sizes are: 500 firms for the Foreign-owned stratum (achieved power 0.872), 700 firms for medium- and large-sized Italian Groups (0.828), and 1,300 firms for Italian SMEs (0.837). Aggregated across the three strata, the design requires approximately 2,500 firms in total to achieve adequate power on the full sample.

7.4 Pooled and Stratum-Specific Inference

A practical implication of the stratum-specific calibration above is that the power analysis supports two distinct inferential strategies, and the choice between them has direct consequences for sampling cost.

The first strategy is to power the full design at 2,500 firms and report a pooled average treatment effect across the entire chemical sector. This is the natural primary specification given our research question, which is framed at the sector level rather than at the level of any single firm type.

The second strategy is to power the design separately within a single stratum and report the treatment effect for that stratum alone. This approach is statistically valid because each stratum's power curve is computed independently using the simulated lognormal distribution specific to that group — the 1,300-firm threshold for SMEs, for instance, holds whether or not the other two strata are included in the experiment.

This second option is operationally relevant because the EUA voucher is a costly intervention, and a phased rollout that begins with a single stratum may be preferable to a full-sample launch. In this regard, the Italian SME stratum is the most natural candidate for a first phase. SMEs account for 40% of total chemical-sector production value and are the most numerous group in absolute terms, making them politically and economically central to the broader policy debate over ETS reform; at the same time, the average voucher cost per SME is substantially lower than for Foreign-owned multinationals, since SMEs have lower baseline import volumes and therefore lower ETS-attributable compliance costs. A phase-one experiment confined to the SME stratum would require 1,300 firms — a large but feasible target — and would yield internally valid causal estimates for this segment without committing the research budget to the full 2,500-firm design upfront. Should the phase-one results justify expansion, the remaining strata could be added in subsequent waves with no loss of statistical validity, since the stratum-specific power curves are independent of one another.

8 Risks and Limitations

This study is designed to estimate whether targeted EU ETS cost relief changes the sourcing behavior of Italian chemical SMEs. The unit of randomization is the firm, while the main outcomes are measured at the firm level using import value, import volume, and the sourcing composition of HS 28–29 intermediate chemical goods. The research design deliberately focuses on SMEs for both policy and practical reasons. From a policy perspective, SMEs are more likely to face binding cost constraints and have less capacity to absorb ETS-related price volatility. From an operational perspective, focusing on SMEs also improves the feasibility of the experiment, particularly in terms of the cost of providing EUA vouchers. However, this SME-focused design also creates several internal validity concerns, external validity limitations, and potential spillover risks.

8.1 Internal Validity

The first internal validity concern is imperfect randomization. Since the experiment is conducted at the firm level, treated and control SMEs must be comparable before the voucher is assigned. If treatment firms differ systematically from control firms in baseline characteristics such as procurement scale, import dependence, sourcing composition, HS subsector, geographic location, or ETS cost exposure, post-treatment differences in sourcing behavior may reflect these pre-existing differences rather than the causal effect of the emissions credit. To reduce this risk, the study will conduct baseline balance checks on key pre-treatment characteristics, including procurement scale, import dependence, sourcing composition, HS subsector, geographic location, and ETS cost exposure. Because the study is restricted to Italian SMEs, assignment will not be stratified by firm type. If small baseline imbalances remain after randomization, the main regression will include pre-treatment covariates to improve precision.

A second concern is imperfect compliance with the treatment. Firms assigned to receive the EUA voucher may not use it as anticipated by the research design. They may use it for compliance, bank it, sell it, or fail to incorporate it into their sourcing decisions. For this reason, the main estimate should be interpreted as an intention-to-treat effect: the effect of being offered the emissions credit, rather than the treatment-on-the-treated effect of actually using the credit.

Third, treatment mechanism ambiguity remains a major limitation. Even if treated SMEs reduce the share or value of imports from third countries, this does not automatically prove that the voucher worked through the intended mechanism of making EU-based inputs relatively more attractive. The voucher may also affect firm behavior through other channels, including cash flow, production expansion, product mix, allowance banking, or allowance sales. To address this concern, the experiment will collect intermediate outcomes in addition to final sourcing outcomes. These include whether the voucher was used for compliance, whether EUA surrender changed, whether the firm banked or sold allowances, whether production volume changed, whether emissions changed, and whether the firm switched suppliers.

Fourth, differential attrition may threaten comparability between treated and control firms. Some SMEs may drop out before the endline survey, exit the market, or stop reporting procurement information during the study period. If attrition differs systematically between treatment and control groups, the follow-up sample may no longer preserve the original randomized comparison. This is especially relevant for SMEs because they may be more financially fragile than larger firms. The study will therefore report attrition rates by treatment status, compare the baseline characteristics of firms that remain with those that drop out, and test whether attrition is correlated with baseline import dependence or ETS exposure.

Fifth, experimental effects may alter firm behavior independently of the voucher. Treated SMEs may reduce third-country sourcing because they know they are part of a carbon-related pilot, while control firms may change their behavior to demonstrate that they would also qualify for future support. These Hawthorne or John Henry effects could reduce the credibility of the estimated treatment effect. To mitigate this risk, the study's public description and survey design should avoid framing the experiment too explicitly around "carbon leakage" or implying that one sourcing pattern is expected or preferred.

Sixth, measurement error is a potential concern because the outcomes rely partly on self-reported firm-level import value, import volume, and sourcing shares. Firms may misclassify HS 28–29 inputs, report supplier regions imprecisely, confuse intermediate inputs with final goods, or provide inconsistent procurement figures across survey waves. To reduce these risks, the study will use data validation procedures, including HS code-description checks, value-to-quantity magnitude checks, cross-field consistency rules, and, where possible, validation against customs records or administrative trade data.

8.2 External Validity

This study deliberately focuses on Italian chemical SMEs. Therefore, the results should be interpreted as evidence on how trade-exposed Italian chemical SMEs respond to targeted ETS cost relief, rather than as evidence for the entire Italian chemical sector. This focus is substantively appropriate because SMEs are central to the Italian chemical industry and are more likely to face binding cost constraints. Compared with larger firms, SMEs may have less access to capital, weaker bargaining power with suppliers, and less flexibility to absorb carbon-cost volatility. However, this design also limits generalizability to large firms, which may have more diversified supplier networks, stronger compliance capacity, greater access to finance, and more flexibility to absorb or pass through carbon-related costs.

The study is also limited to active SMEs with observable procurement or trade data. This means the findings may not generalize to very small chemical firms, firms without meaningful import activity, firms that rely primarily on domestic suppliers, or firms with incomplete procurement records. This restriction is necessary for measurement and statistical power, but it narrows the population to which the results apply.

Sectoral and geographic specificity also limit external validity. The study focuses on Italian firms producing or importing HS 28–29 chemical intermediates. These products have specific supply-chain structures and particular exposure to EU ETS and CBAM policy. Therefore, the results may not generalize to other sectors, such as steel, cement, aluminum, or energy-intensive manufacturing more broadly. The results may also not generalize to chemical SMEs in other EU countries, where energy prices, industrial policy support, domestic supplier capacity, and procurement networks may differ substantially from those in Italy.

Another external validity limitation concerns the treatment itself. The intervention is a temporary EUA voucher, not a permanent suspension of the EU ETS or a long-term free-allocation scheme. SMEs may be reluctant to restructure supply chains in response to temporary relief, especially if supplier switching involves quality risks, certification requirements, contract renegotiation, or long-term relationship costs. Therefore, the study estimates the short-term sourcing response to targeted ETS cost relief and should not be interpreted as a direct indicator of the long-term effects of broader ETS reform.

Finally, the experiment estimates a partial-equilibrium firm-level effect. If the voucher were scaled to all chemical SMEs or to the broader chemical sector, market-wide effects could emerge. Higher demand for EU-based inputs could increase supplier prices or tighten supplier capacity, reducing firms' ability to shift away from third-country suppliers. Conversely, sustained demand from SMEs could encourage EU suppliers to expand capacity, improve reliability, or lower unit costs, which could also benefit untreated firms. The experimental estimate should therefore be interpreted as the effect of a targeted voucher within a controlled pilot, not as a complete estimate of the economy-wide effect of ETS reform.

8.3 Spillovers and Other Threats

There are several spillover channels that may complicate interpretation. Because treated and control SMEs operate in the same sector and may rely on overlapping supplier networks, treated firms' increased demand for EU-based inputs could affect supplier prices or the capacity available to control firms. Information may also spill over if treated firms share new sourcing options or compliance strategies through industry networks. In addition, treated firms may use lower compliance costs to reduce prices or expand output, indirectly affecting control firms' sourcing and market behavior. Finally, because the treatment involves EUA vouchers, firms may bank, sell, or trade allowances, creating possible allowance-market spillovers.

There are two further limitations. First, sourcing shifts are only a proxy for carbon leakage risk. A decline in imports from third countries would suggest that ETS cost relief affects sourcing behavior, but it would not directly prove that global emissions have fallen. Stronger claims would require supplier-level lifecycle emissions estimates, which are not conventionally available. Second, the policy may face legal and political constraints, since randomly allocating EUA vouchers to selected SMEs could raise concerns under EU ETS governance rules, state-aid rules, and broader debates over whether carbon-pricing relief weakens climate policy.

Overall, the experimental design mitigates many internal validity concerns through random assignment, baseline comparability checks, pre-treatment covariate adjustment, and a conservative intention-to-treat interpretation. The remaining limitations mainly reflect the intended scope of the study rather than flaws in the research design. In sum, the results should be interpreted as a targeted test of whether ETS cost relief changes import value, import volume, and sourcing composition among Italian chemical SMEs, rather than as a full estimate of broader ETS reform, the chemical sector as a whole, or economy-wide carbon leakage.

9 Conclusion

This Pre-Analysis Plan establishes a framework to assess the effectiveness of EU ETS EUAs in reducing emissions by mitigating compliance costs. Through a Randomized Control Trial, the study provides targeted emissions credits to examine if financial relief permits firms to source domestically, instead of from unregulated “Third Country” suppliers. Should the null hypotheses be rejected, we should see firms altering sourcing from third countries, ultimately increasing domestic emissions.

As the European chemical industry faces increasing pressure from high prices and increasing regulations, this research provides timely evidence on the tension between decarbonization objectives and industrial competitiveness. By focusing on SMEs, the research offers specific insights into a vulnerable economic segment, which comprises a significant share of value added, but lacks the capacity of larger groups to absorb price volatility.

To build on this research, several areas warrant future investigation as the regulatory landscape evolves:

1) Stricter CBAM Reductions: As the Linear Reduction Factor increases, future studies should re-examine if sourcing behavior shifts at higher reduction levels with more expensive vouchers.
2) Inclusion of Multinationals and medium- and large-sized Italian firms: This study emphasizes SMEs, which account for the largest share of production value. However, multinationals and larger firms can better navigate price distortions and may exhibit different sourcing behavior. They are omitted from this study due to cost constraints and the inequitable nature of Italian taxpayers subsidizing foreign firms. As evidenced in the power calculations, however, there is sufficient information for further analysis.

By committing to this transparent and pre-specified methodology, the study ensures that its results—whether they show a significant shift in sourcing or a null effect—will provide high-value data to inform the future of EU climate economics and industrial policy. While we acknowledge this study may increase emissions, the information gained will inform the effectiveness of vouchers in net global emissions and provide insight for future environmental regulation.

10 Bibliography

Banca d'Italia. (2023, June 30). *Indagine sulle imprese industriali e dei servizi nell'anno 2022* [Survey of industrial and service firms in 2022]. https://www.bancaditalia.it/pubblicazioni/indagine-imprese/2022-indagini-imprese/index.html.

Böning, Justus, Virginia di Nino, and Till Folger. "Benefits and Costs of the ETS in the EU, a Lesson Learned for the CBAM Design." ECB Working Paper No. 2023/2764

Cefic. "Landscape of the European Chemical Industry: Italy." Retrieved April 6, 2026. https://cefic.org/landscape-of-the-industry/italy/.

Climatiq. "How to calculate emissions for CBAM reporting." Published December 29, 2025. Retrieved April 1, 2026. https://www.climatiq.io/blog/how-to-calculate-emissions-for-cbam-reporting.

Colmer, Jonathan, Ralf Martin, Mirabelle Muuls, and Ulrich Wagner. "Does Pricing Carbon Mitigate Climate Change? Firm-Level Evidence from the European Union Emissions Trading System." *The Review of Economic Studies* (2024).

Coster, Pierre, Julian di Giovanni, and Isabelle Mejean. "Firms' Supply Chain Adaptation to Carbon Taxes." *Federal Reserve Bank of New York Staff Reports* 1136 (2025).

David, Sébastien, Christian Popa, Gunter Lipowsky, Paul Petrescu et al. (Advancy). The Competitiveness of the European Chemical Industry. CEFIC, January 2025. https://cefic.org/app/uploads/2025/05/Cefic-Advancy-study-The-Competitiveness-of-the-European-Chemical-Industry.pdf

Deganello, Sara. "Economic Impact of the ETS System on Italian Industry: Costs, Sectors and Requests for Revision." *Il Sole 24 Ore.* March 16, 2026. https://en.ilsole24ore.com/art/ets-how-much-does-mechanism-weigh-on-italian-industry-AIXmeFvB

Dell'Omodarme, Giorgio. "Clash over ETS ahead of European Council: eight EU governments oppose suspension proposal." Eunews.com, March 12, 2026. https://www.eunews.it/en/2026/03/12/clash-over-ets-ahead-of-european-council-eight-eu-governments-oppose-suspension-proposal/.

Echemi. "How did the European chemical industry give birth to giants such as BASF/Evonik?" July 13, 2020.

Elderson, Frank. "Europe's fossil fuel dependence poses risks to price stability." European Central Bank, *The ECB Blog* (blog), April 7, 2026.

EU Emissions Tracker. *Chemical Industry*. Retrieved May 01, 2026. https://emission-tracker.eu/italy/industry/chemical-industry.

European Commission. (n.d.) *About the EU ETS.* Retrieved March 25, 2026. https://climate.ec.europa.eu/eu-action/carbon-markets/about-eu-ets_en.

European Commission. *CBAM Sectors*. Retrieved April 25, 2026. https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism/cbam-sectors_en

European Commission. "EU Emissions Trading System Has Reduced Emissions in the Sectors Covered by 50% Since 2005." April 4, 2025.

European Committee of the Regions. *A Place-Based Approach to Strengthening the EU Chemicals Industry in Regions and Cities*. Commission for Economic Policy, 2026.

European Environment Agency. “Greenhouse Gas Emissions under the EU Emissions Trading System.” November 6, 2025. https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-under-the.

European Environment Agency. “Total Greenhouse Gas Emissions in the Chemical Industry (Indicator).” May 7, 2025. https://www.eea.europa.eu/en/european-zero-pollution-dashboards/indicators/total-greenhouse-gas-emissions-in-the-chemical-industry.

Eurostat. (May 13, 2025). Number of chemicals and chemical products manufacturing firms in Italy from 2021 to 2024 (in 1,000s). In *Statista*. Retrieved April 27, 2026, from https://www.statista.com/statistics/375238/number-of-enterprises-in-the-manufacture-of-chemicals-sector-in-italy/

Federchimica Confindustria. *The Role of Foreign-Owned Chemical Companies in Italy*. https://www.federchimica.it/docs/default-source/chemical-industry-in-italy/the-chemical-industry-in-italy-2019_web.pdf

Federchimica Confindustria. *The Chemical Industry in Italy.* 2025. https://www.federchimica.it/docs/default-source/chemical-industry-in-italy/the-chemical-industry-in-italy-2019_web.pdf

Federico, S., & Tosti, E. (2017). Exporters and importers of services: Firm-level evidence on Italy. *The World Economy*, 40(10), 2078–2096. https://doi.org/10.1111/twec.12462.

Grimes, Sean. "China's Evolving Role in the Chemical Global Value Chain." *The Chinese Economy* 56 no. 6 (2023): 441–58. https://www.tandfonline.com/doi/full/10.1080/10971475.2023.2213631

ICN Bureau. “Europe Faces Chemical Industry Collapse As Chinese Imports Flood Market: Report.” *Indian Chemical News*, December 19, 2025.

International Carbon Action Partnership. “EU Emissions Trading System (EU ETS).” Retrieved April 25, 2026. https://icapcarbonaction.com/en/ets/eu-emissions-trading-system-eu-ets

Kuppel, Anna. “CBAM Timeline, Deadlines and Phases: What to Expect in 2026.” Coolset, January 4, 2026. https://www.coolset.com/academy/cbam-timeline-deadlines-phases-what-to-expect-2026

Minten, Hannah, Julian Hausweiler, Benedict Probst, Christiane Reinert, Raoul Meys, Andre Bardow. “Embodied Emissions of Chemicals within the EU Carbon Border Adjustment Mechanism.” *Nature Sustainability* 8 (2025): 1381–1390. https://www.nature.com/articles/s41893-025-01618-5

Myllyvirta, Lauri. “Analysis: China’s CO2 emissions have now been ‘flat or falling’ for 21 months.” *CarbonBrief,* February 12, 2026.

Naegele, Helene, and Aleksandar Zaklan. “Does the EU ETS Cause Carbon Leakage in European Manufacturing?” *Journal of Environmental Economics and Management* 93, (2019): 125–47.

Offset8 Capital. “EU Carbon Market 2025: Emissions Trading System Developments & Business Impacts.” September 15, 2025. https://offset8capital.com/articles/eu-carbon-market-2025-emissions-trading-system-developments-business-impacts/

Pacheco, Marta. 2026. “Italy calls for suspension of EU carbon market.” Euronews.com. https://www.euronews.com/my-europe/2026/02/26/italy-calls-for-suspension-of-eu-carbon-market

Roland Berger. *European Chemical Closures and Investments Radar 2022–2025*. Independent Radar reports, Cefic, December 2025.

Senken. “EU ETS.” Updated February 1, 2026. https://www.senken.io/glossary/eu-ets

Thompson, Charlie. “The Carbon Border Adjustment Mechanism (CBAM): Overview and Timeline.” Glassdome (blog), January 8, 2025. https://glassdome.com/cbam-overview-and-timeline/

Toshev, Atanas, Anna A. Mihaylova, Vasil Madzharov, Elina Petkova-Gueorguieva, Bogdan Kirilov, Dimitar Mirchev, and Stanislav Gueorguiev. “The European Union’s Dependence on Imports of Active Pharmaceutical Ingredients from Third Countries—Challenges and EU Initiatives.” *Pharmacia* 72 (2025): 1-6. https://pharmacia.pensoft.net/article/172383/

Trading Economics. “EU Carbon Permits.” Retrieved April 25, 2026. https://tradingeconomics.com/commodity/carbon

United Nations. (2026). *UN Comtrade database: HS Chapters 28 and 29* [Data set]. https://comtradeplus.un.org/TradeFlow.

Vangenechten, Domien, Vilislava Ivanova, and Rheanna Johnston. “Facts over Fiction: Why the EU ETS Is Key for a Competitive, Secure Europe.” *E3G,* March 16, 2026. http://e3g.org/publications/facts-over-fiction-why-the-eu-ets-is-key-for-a-competitive-secure-europe/.

Verde, Stefano F. “The Impact of the EU Emissions Trading System on Competitiveness and Carbon Leakage: The Econometric Evidence.” *Journal of Economic Surveys* 34 (2020): 320-343.

Volza. (2026, March 23). *Italy import trade data* [Data set]. https://www.volza.com/global-trade-data/italy-import-trade-data/.

World Integrated Trade Solution. “Italy 2023 Import Partner Share.” https://wits.worldbank.org/CountryProfile/en/Country/ITA/Year/LTST/TradeFlow/Import/Partner/by-country/Product/28-38_Chemicals.